

Germinal Center Dynamics Revealed by Multiphoton Microscopy with a Photoactivatable Fluorescent Reporter

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SUMMARY

The germinal center (GC) reaction produces high-affinity antibodies by random mutation and selective clonal expansion of B cells with high-affinity receptors. The mechanism by which B cells are selected remains unclear, as does the role of the two anatomically defined areas of the GC, light zone (LZ) and dark zone (DZ). We combined a transgenic photoactivatable fluorescent protein tracer with multiphoton laser-scanning microscopy and flow cytometry to examine anatomically defined LZ and DZ B cells and GC selection. We find that B cell division is restricted to the DZ, with a net vector of B cell movement from the DZ to the LZ. The decision to return to the DZ and undergo clonal expansion is controlled by T helper cells in the GC LZ, which discern between LZ B cells based on the amount of antigen captured and presented. Thus, T cell help, and not direct competition for antigen, is the limiting factor in GC selection.

INTRODUCTION

Germinal centers (GCs) were first described in the 19th century as distinct microanatomical regions in lymphoid organs that contained dividing cells and were long believed to be the sites of lymphocyte development (Nieuwenhuis and Opstelten, 1984). It has since become evident that, in fact, GCs are the site of antigen-dependent clonal expansion, immunoglobulin diversification, and affinity maturation (Allen et al., 2007a; Klein and Dalla-Favera, 2008; MacLennan, 1994; Rajewsky, 1996; Tarlinton, 2008), all of which are required for the generation of the high-affinity antibodies that make up the core of the humoral immune response.

Affinity maturation is defined as the gradual increase in the affinity of serum antibodies following infection or immunization (Eisen and Siskind, 1964; Goidl et al., 1968; Nussenzweig and Benacerraf, 1967). This process occurs in the GC as the result of random somatic hypermutation of B cell receptor (BCR) genes (McKean et al., 1984) followed by Darwinian-like selection of B cell clones with increased affinity for antigen (Allen et al., 2007a; Klein and Dalla-Favera, 2008; MacLennan, 1994; Rajewsky, 1996; Tarlinton, 2008). Despite the importance of affinity-based selection, there is little understanding of the mechanisms by which this process is controlled within the GC.

Early histological studies advanced the idea that the GC is divided into two anatomically distinct regions: a dark zone (DZ), containing large, mitotically active B cells known as *centroblasts*; and a light zone (LZ), containing smaller, nondividing B cells known as *centrocytes*, as well as antigen deposited on the surface of follicular dendritic cells (FDCs) and antigen-specific follicular T helper cells (Allen et al., 2007a; Klein and Dalla-Favera, 2008; MacLennan, 1994; Nieuwenhuis and Opstelten, 1984; Rajewsky, 1996; Tarlinton, 2008). This segregation between cell division and potential selecting agents suggested a model in which selection would require the migration of cells between the two zones, the DZ acting as a source of B cells with mutated BCRs that would then undergo selection in the LZ (MacLennan, 1994), possibly returning to the DZ for further proliferation and mutation (Meyer-Hermann et al., 2001; Oprea and Perelson, 1997).

Recent live-imaging studies demonstrated that B cells do indeed move bidirectionally between the two GC zones (Allen et al., 2007b; Hauser et al., 2007a; Schwickert et al., 2007). However, the length of the imaging window and the number of interzonal migration events documented were insufficient to draw firm conclusions regarding the dynamics of GC selection (Allen et al., 2007a; Figge et al., 2008; Meyer-Hermann et al., 2009). Furthermore, GC B cells in the two zones appeared similar in size and morphology and showed similar levels of DNA synthesis, challenging the traditional description of small,

nondividing centrocytes and large, cycling centroblasts (Allen et al., 2007b; Hauser et al., 2007a; Schwickert et al., 2007). Finally, studies in which zonal migration was disrupted failed to show alterations in cell division, GC size, or development of high-affinity anti-NP antibodies (Allen et al., 2004; Nie et al., 2004). Together with the initial short-term imaging studies, these physiologic experiments called into question the importance of polarization of the GC into LZ and DZ in terms of selection.

Another key unresolved issue is precisely how antigen-driven selection is regulated. Traditionally, two nonmutually exclusive possibilities have been considered based on the signals available to B cells in the GC. According to the classical model, B cells with higher-affinity receptors would be selected as a result of BCR crosslinking by antigen deposited as immune complexes on the FDC surface (Tarlinton and Smith, 2000). A second possibility is that GC B cells use their BCRs to capture and internalize such antigen for presentation to GC-resident T cells. These specialized T helper cells would in turn influence B cell selection through cytokine secretion or coreceptor ligation (Allen et al., 2007a; Tarlinton, 2008; Vinuesa et al., 2005b). Despite the wealth of studies addressing this question, the available evidence is both circumstantial and contradictory, and models of the way in which selection takes place in the GC remain speculative (Allen et al., 2007a; Hauser et al., 2007b; Meyer-Hermann et al., 2006; Tarlinton, 2008).

Solving the issue of how GC selection operates would require a precise definition of the characteristics of LZ and DZ B cells and of the migration patterns of these cell populations in vivo. To this end, we developed a method for in situ microanatomical labeling and long-term in vivo imaging of GC B cells that combines a new transgenic mouse that expresses photoactivatable green fluorescent protein (PA-GFP) (Patterson and Lippincott-Schwartz, 2002), multiphoton laser scanning microscopy, and flow cytometry. Here, we report on the characteristics of LZ and DZ GC B cells and on the dynamic mechanisms that limit interzonal migration and affinity-based selection during the humoral immune response.

RESULTS

Photoactivation

PA-GFP is a green fluorescent protein variant whose peak excitation wavelength shifts from ~415 nm to ~495 nm upon one-photon irradiation at ~415 nm or two-photon irradiation at ~720–840 nm (Patterson and Lippincott-Schwartz, 2002; Schneider et al., 2005). To examine selection in the GC, we produced transgenic mice in which all hematopoietic cells express PA-GFP (Figures S1A–S1C available online). PA-GFP-expressing cells can be photoactivated within intact lymph nodes with great microanatomical precision (~10 microns in the Z dimension, or close to one cell diameter; Figure 1A) by two-photon irradiation at 830 nm and subsequently identified by two-photon excitation at 940 nm (Movie S1) or flow cytometry using a conventional 488 nm laser (Figure 1B). After a brief recovery period, the migration of photoactivated naive T and GC B cells is indistinguishable from that of control cells (Figures 1C and 1D, Movie S2, and below). The half-life of photoactivated PA-GFP in naive B cells was estimated to be 30 hr (Figure S1D),

a figure consistent with previous estimates for the half-life of GFP in living cells (Corish and Tyler-Smith, 1999; Nagaoka et al., 2000).

Photoactivatable Germinal Centers

Although two populations of GC B cells were originally defined based on size, LZ and DZ B cells were found to be indistinguishable in size or movement pattern by multiphoton microscopy (Allen et al., 2007b; Hauser et al., 2007a; Schwickert et al., 2007), and there are no anatomically validated surface markers for the two cell types. To mark LZ and DZ cells directly in situ, we produced photoactivatable antigen-specific GCs. PA-GFP mice were bred to B1-8^{hi} heavy chain knockin mice (PA-GFP/B1-8^{hi}), which carry a heavy chain that, when combined with an Igλ light chain, produces an antibody that is specific for the hapten 4-hydroxy-3-nitrophenylacetyl (NP) (Shih et al., 2002). To generate photoactivatable GCs, we used an adaptation of the classical prime-boost model (Katz et al., 1970; Rajewsky et al., 1969; Schwickert et al., 2007). PA-GFP/B1-8^{hi} B cells were transferred into ovalbumin (OVA)-primed mice, which were subsequently challenged subcutaneously with soluble NP-conjugated OVA to produce NP-specific GCs in skin-draining lymph nodes (Figure 2A). This protocol is preferable for intravital imaging due to the absence of adjuvant-induced inflammation in the draining lymph node. The GC light zone was identified by injecting NP conjugated to red fluorescent protein tdTomato (Shaner et al., 2004) (NP-tdTomato), which forms immune complexes that bind to FDCs in NP-immunized mice (Fooksman et al., 2010).

LZ and DZ B Cells

To determine the phenotype of LZ and DZ B cells, we photoactivated one or the other zone in explanted lymph nodes (Figure 2B and Movie S3) and analyzed the photoactivated cells by flow cytometry. LZ and DZ cells expressed comparable levels of FAS, GL-7 antigen, CD38, and MHC II (Figure S2A). Consistent with previous immunohistochemical and functional studies (Allen et al., 2004; Liu et al., 1991), LZ cells expressed higher levels of surface immunoglobulin, slightly higher levels of chemokine receptor CXCR5, and lower levels of CXCR4 than DZ cells (Figure 2C). Cell-cycle analysis by DNA content showed that cells in the G2/M phase of the cell cycle were almost entirely restricted to the DZ, whereas cells in S phase were found in both zones (Figure 2D). Contrary to the classical description of centrocytes and centroblasts, LZ and DZ B cells were similar in size and were both larger than follicular B cells (Figure 2E). An exception to this was a small subpopulation of larger DZ cells (5%–10%) that corresponds to cells in the G2/M phase of the cell cycle (Figure S2B).

To further characterize LZ and DZ B cells, we purified photoactivated cells by cell sorting (Figure 2A) and performed microarray-based gene expression profiling. Only 144 of approximately 45,000 probes reproducibly differed by more than 2-fold between the two compartments (Figures 3A and 3B, Table S1). By contrast, comparison of gene expression by either LZ or DZ cells and naive B cells showed a substantially larger difference, with 2135 and 2458 probes differing by 2-fold or more, respectively (Figure 3A). Systematic Gene Ontology analysis of an

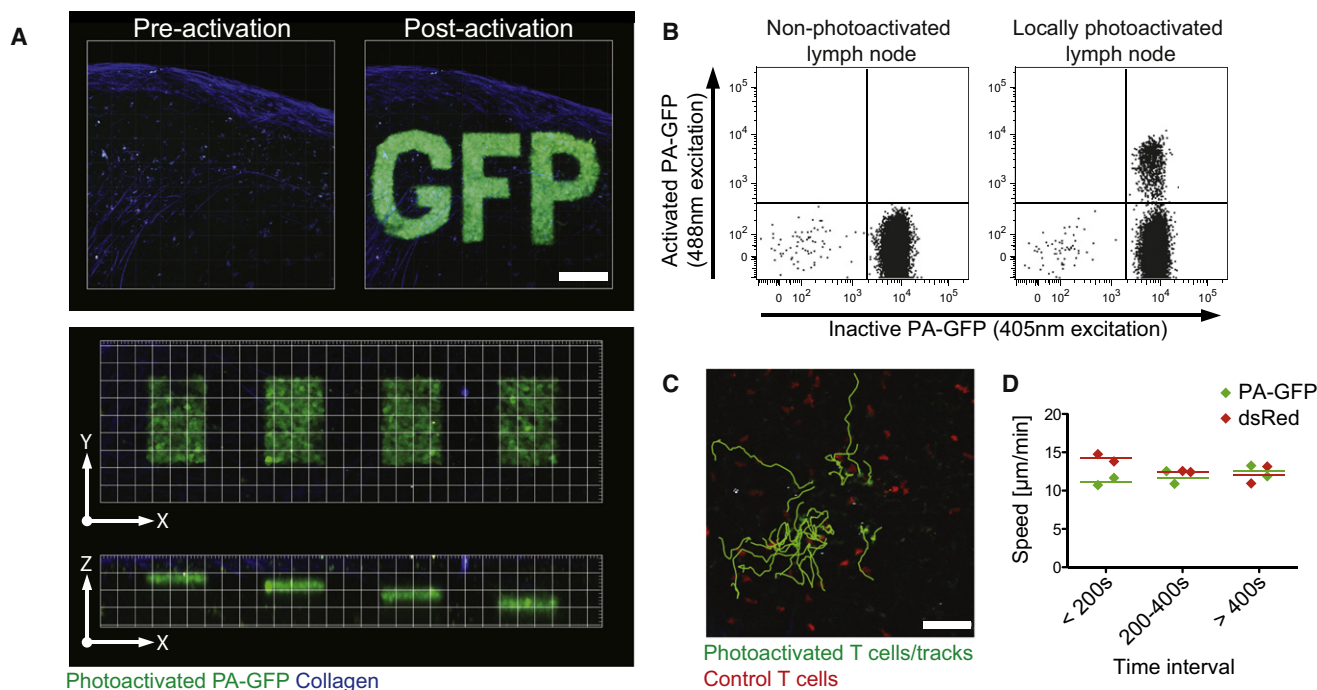


Figure 1. Microanatomical Labeling of PA-GFP Transgenic B Cells

(A) Spatial precision of multiphoton photoactivation of explanted inguinal lymph nodes from PA-GFP transgenic mice. Top: precision in the X-Y axis. Collapsed Z stacks (60 μm , 5 μm steps). Scale bar = 100 μm . Imaged at $\lambda = 940$ nm before and after photoactivation at $\lambda = 830$ nm of a defined region of interest ("GFP"). Bottom: precision in the Z axis. X-Y and X-Z views of single planes photoactivated at different Z positions 10 μm apart, from 30 to 60 μm below the lymph node capsule. Grid = 20 $\times$ 20 μm .

(B) Detection of photoactivated cells by flow cytometry. Nonphotoactivated (left) and locally photoactivated inguinal lymph nodes from a PA-GFP transgenic mouse were processed in parallel and analyzed for fluorescence under 407 nm (inactive PA-GFP) and 488 nm (activated PA-GFP) excitation (emission filter = 530/30 nm for both). Photoactivation as described for (A).

(C and D) Cells photoactivated in vivo remain viable. (C) Tracks of transferred T cells photoactivated in vivo. (D) Mean velocity of photoactivated and control cells at different time intervals after photoactivation. All photoactivated cells plus an equal number of randomly selected control cells (dsRed transgenic) outside the photoactivation area were tracked. Each symbol represents one independent experiment (1 movie per experiment, $\sim$ 20 cells of each type per movie).

expanded subset of genes whose expression differed between LZ and DZ by at least 1.33-fold (see [Extended Experimental Procedures](#)) showed that probes upregulated in DZ were dominated by genes involved in mitosis, whereas probes upregulated in LZ were enriched in lymphocyte activation markers, cell-surface molecules, and regulators of apoptosis (Figures 3B and 3C, Table S2). To gain insight into the pathways activated in each zone, we conducted gene set enrichment analysis (GSEA) (Mootha et al., 2003; Subramanian et al., 2005) for transcription factor motifs (Xie et al., 2005). This analysis showed strong evidence of NF- κ B activation in the LZ, in addition to the signatures of serum response factor (SRF), signal transducer and activator of transcription 5 (STAT5), and c-Myc engagement (Table S3). Because both BCR crosslinking and T cell signals delivered via CD40 are capable of activating the NF- κ B pathway in LZ, we conducted a further GSEA overlaying gene signatures of CD40 and BCR ligation (Basso et al., 2004; Zhu et al., 2004). This analysis showed the LZ population to be strongly imprinted with the signatures of both of these activation signals (Figure 3D and Table S3). Taken together, our cell-cycle and gene expression analyses support the notion that cells in the two zones differ in a functionally significant manner.

To identify markers capable of distinguishing DZ and LZ cells by flow cytometry, we screened photoactivated LZ or DZ B cells on the basis of the gene array results for differentially expressed surface molecules. Two markers, CD86 and CD83, were able to distinguish between LZ and DZ cells by flow cytometry when combined with CXCR4: LZ cells are CXCR4^{lo} CD86^{hi}CD83^{hi} and DZ cells are CXCR4^{hi}CD86^{lo}CD83^{lo} (Figures 4A and 4B). To validate the gene array results obtained with B1-8^{hi} NP-specific B cells, we examined polyclonal LZ and DZ GC cells from immunized wild-type mice identified using these markers (Figure S3A). Flow cytometric analysis (Figure 4C and Figure S3B) or cell sorting followed by qPCR (Figure S3C and Figure 4D) confirmed the differential expression in LZ and DZ of all 23 genes tested. Furthermore, LZ and DZ cells in these mice showed the same cell-cycle distribution as those identified by photoactivation (Figure 4E and Figure S3D). Therefore, LZ and DZ B cells in polyclonal GCs from wild-type mice are phenotypically indistinguishable from their B1-8^{hi} counterparts. We conclude that the combination of photoactivation and flow cytometry allows for unambiguous identification of LZ and DZ GC B cells based on microanatomical location. The LZ contains cells whose gene expression profile is consistent with antigen

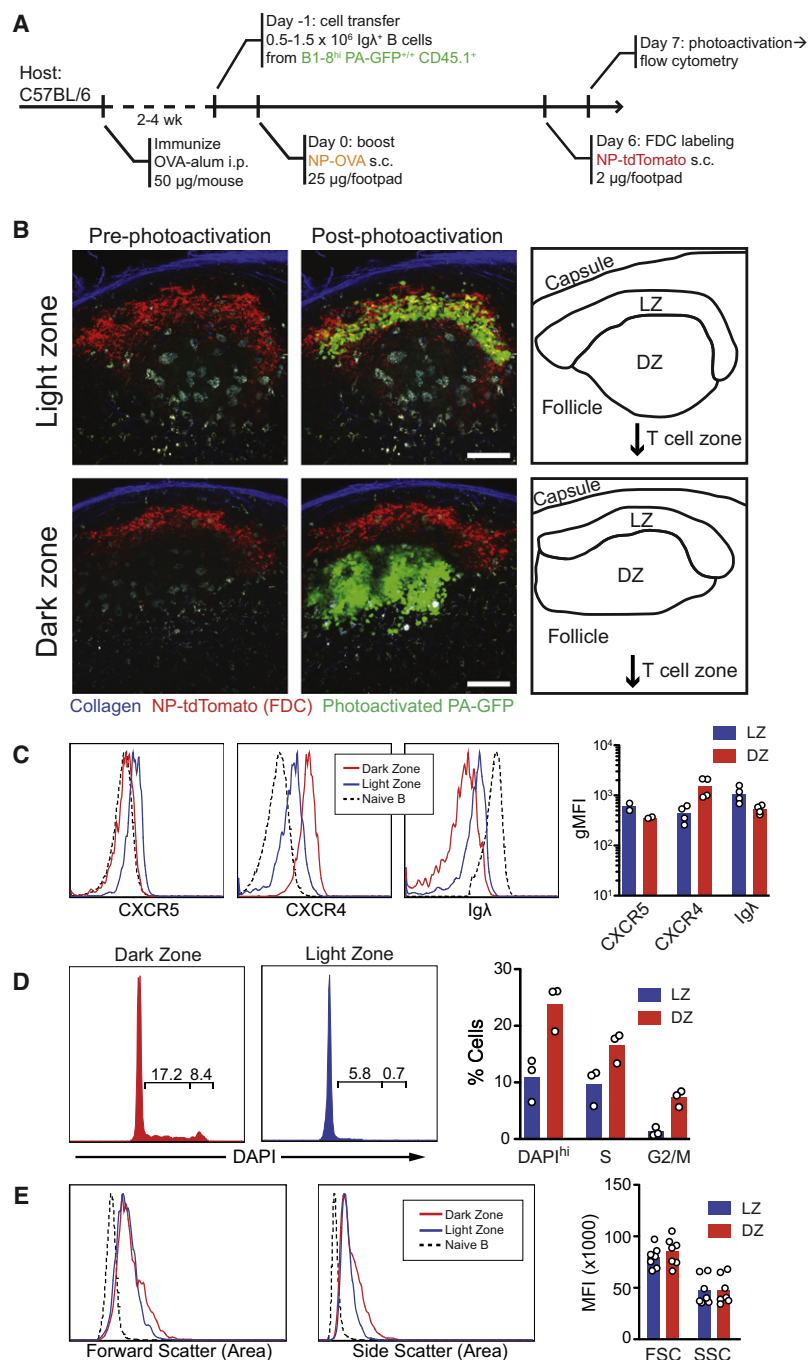


Figure 2. Phenotype of LZ and DZ Cells

(A) Diagrammatic representation of the experimental protocol.

(B) Examples of LZ (top row) and DZ (bottom row) multi-photon photoactivation. Collapsed Z stacks (40 µm, 5 µm steps) are shown. Scale bar = 50 µm. Imaged at $\lambda = 940$ nm before and after photoactivation at $\lambda = 830$ nm. (C–E) Chemokine receptor and BCR expression, DNA content (DAPI), and forward and side scatter in photoactivated LZ or DZ cells. GC cells gated as live/single, CD45.1⁺, FAS^{hi}. Non-GC cells (shown for comparison) gated as IgA⁺FAS⁺. Left panels: quantification of flow cytometry data. Each symbol represents one experiment (pools of 3–5 mice per experiment).

analyses by comparing their motility to that of control CFP⁺ GC cells in the same sample. Photoactivated LZ and DZ cells were similar to control cells in terms of morphology and speed of movement for up to 8 hr after photoactivation (Figure 5E and Movie S4 and Movie S5).

Following photoactivation, DZ B cells migrated rapidly to the LZ, with up to 50% of cells reaching the opposite zone by 4 hr (Figures 5B and 5D). In stark contrast, photoactivated LZ B cells crossed into the DZ at a much lower rate, with less than 15% making the transition over a 6 hr period (Figures 5C and 5D). Similar results were obtained when B-Phycoerythrin (PE) immune complexes were used to label the LZ instead of NP-tdTomato (Allen et al., 2007b) (Figure 5D). Thus, there is a strong net flux of cells from DZ to LZ, with only a few cells returning from LZ to DZ. Importantly, measurements made 30 min after photoactivation showed only a minor difference in the percentage of cells migrating from DZ to LZ (5.7%, $\pm 2.2\%$) and LZ to DZ (3.7%, $\pm 0.7\%$), potentially accounting for the inability of prior studies to detect a net vector in interzonal migration.

To determine what these rates of migration imply in terms of selection, we modeled the interzonal migration data using an ordinary differential equation model for the dynamics of photoactivated B cells in GC zones (see Extended Experimental Procedures for details). This approach yielded migration rates of 15%/hr for DZ to LZ and 3%/hr for LZ to DZ,

encounter and T-B cell interaction, whereas cells in the DZ are poised to undergo cell division.

Interzonal Migration

To examine the exchange of GC B cells between zones over an extended period of time, we photoactivated NP-specific LZ or DZ cells in popliteal lymph nodes of living mice and tracked their migration at 1 hr intervals for up to 6 hr (Figures 5A–5D). Photoactivated cells were monitored for viability throughout these

which corresponds to a fraction of LZ cells selected to return to DZ of 30% in this experimental system. The fitting of the simulated data to the experiment is shown in Figure S4. Similar results were obtained using a more complex agent-based model of GC dynamics, morphology, and affinity maturation published previously (not shown) (Figge et al., 2008). Combined with the finding that mitosis is restricted to the DZ, the net flow of cells from the DZ to the LZ strongly suggests that selection is driven by competition among LZ cells for signals capable of

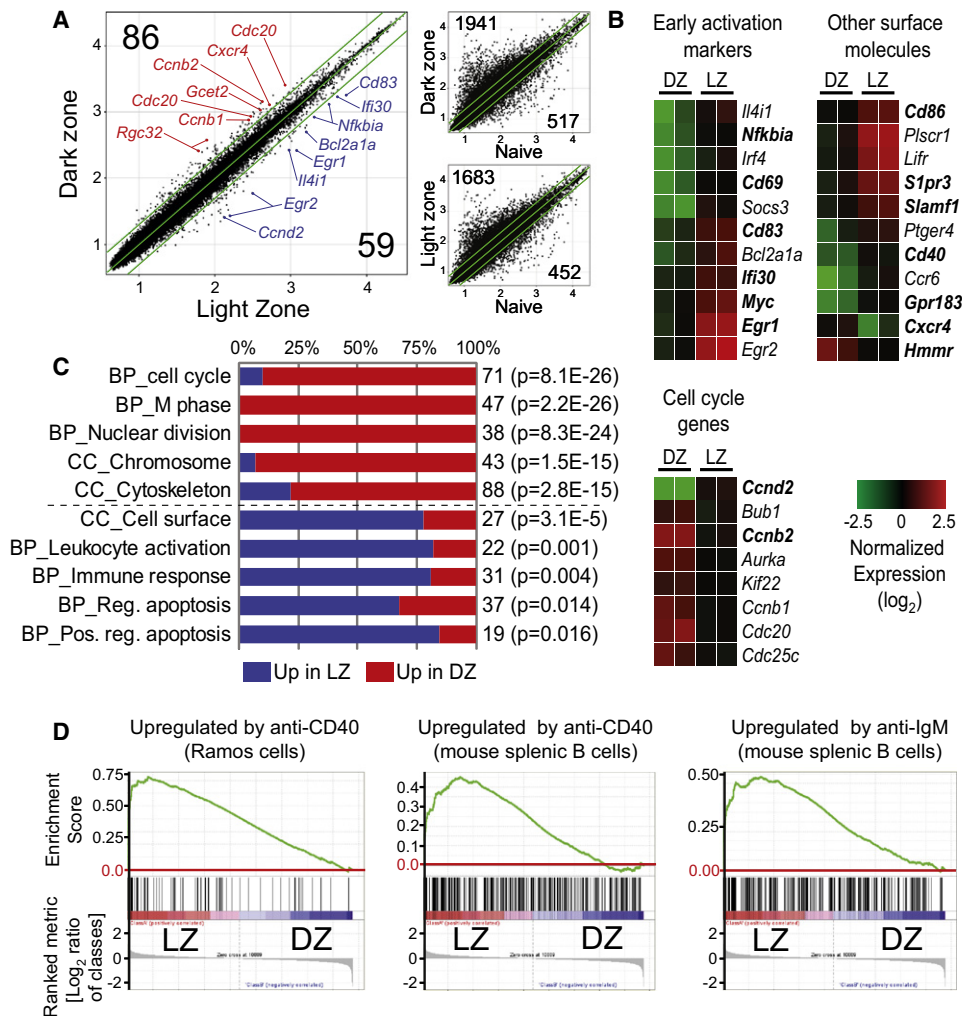


Figure 3. Gene Expression in LZ and DZ Cells

(A) Differences in gene expression between cells photoactivated in LZ and DZ. Samples generated as indicated in Figure 2A and sorted as live/single, CD45.1⁺, FAS^{hi}/GL-7⁺, activated PAGFP⁺. Large panel: scatter plots showing raw expression levels ($\log_{10}$) for ~45,000 microarray probes in LZ and DZ. Selected genes are indicated in red (up in DZ) or blue (up in LZ). Smaller panels: differences in gene expression between LZ or DZ and naive B cells, shown for comparison. Green lines indicate 2-fold boundary. Numbers in black indicate the number of genes upregulated by more than 2-fold in the corresponding zone in both replicate assays. (B) Heatmaps showing normalized expression ($\log_2$) of selected genes whose expression differed by more than 2-fold between LZ and DZ in both replicate assays. The full list of genes in this category can be found in Table S1. Genes in bold type were confirmed by either flow cytometry or qPCR.

(C) Gene Ontology (GO) analysis (biological process [BP] and cellular compartment [CC]) of an expanded list of genes differing by at least 1.33-fold between LZ and DZ (see Extended Experimental Procedures and Table S1). Blue and red bars represent the fraction of genes in a given GO category upregulated in LZ and DZ, respectively. The total number of genes is indicated to the right of the bar, as is the p value (Benjamini correction) for enrichment in the zone with the larger fraction of the genes over the whole mouse genome. An expanded GO analysis is presented in Table S2.

(D) GSEA analysis showing enrichment of gene signatures of CD40 and BCR ligation (gene sets obtained from the literature) in LZ cells. All nominal p values and FDR rates < 0.001. An expanded GSEA analysis is presented in Table S3.

All data are the average of two experiments, each corresponding to a pool of 15–20 lymph nodes per condition.

promoting their return to the DZ for further rounds of proliferation.

Access to T Cell Help Limits Interzonal Migration and GC Selection

To gain insight into which signals may be required for LZ to DZ migration, we initially took advantage of the fact that, in addition to the transferred B1-8^{hi} B cells, GCs in our transfer model

(Figure S5A) also contain a minor population (~10%) of endogenous B cells (Figure S5B), which likely represents a mixture of OVA- and NP-specific cells. This population displays a skewed DZ/LZ ratio (~0.5:1, as opposed to ~2:1 among B1-8^{hi} cells) and a lower proliferation rate, consistent with a lower rate of selection (Figure S5B). To determine whether this is a consequence of the inability of endogenous cells to compete with the B1-8^{hi} population for signals that regulate return to the DZ,

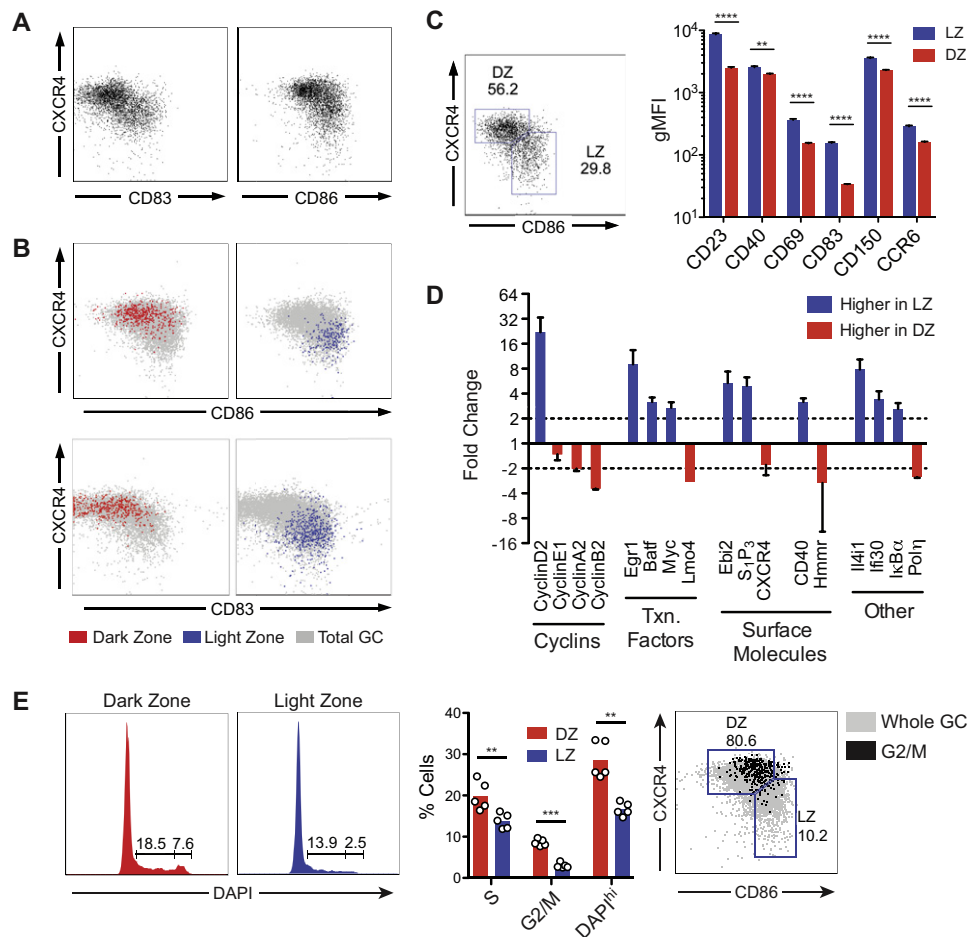


Figure 4. Distinguishing LZ and DZ Cells by Flow Cytometry

(A) Examples of distribution of GC cells according to expression of CXCR4 and CD83 or CD86 in day 7 secondary GCs generated as shown in Figure 2A. Gated on B220⁺CD45.1⁺CD38⁺FAS⁺.

(B) Distribution of GC cells photoactivated in LZ or DZ according to expression of CXCR4 and CD86 or CD83. Gated on CD45.1⁺, FAS^{hi}, photoactivated PA-GFP⁺.

(C) Expression of surface markers in LZ and DZ gates by flow cytometry. Polyclonal GCs were generated in wild-type B6 mice as indicated in Figure S3A, and LZ and DZ populations were defined by expression of CD86 and CXCR4, as shown in the left panel. Right panel: expression of surface markers in LZ and DZ gates. Histograms are presented in Figure S3B. Representative of 2–3 independent experiments with 2–3 mice each. Bars = SEM.

(D) Confirmation of microarray results by qPCR. LZ and DZ cells sorted according to expression of CD86 and CXCR4. GCs were generated as in Figure S3A. Sorting strategy and post-sort purity are shown in Figure S3C. Graph represents data from two experiments with pools of 10 mice each. Bars = range.

(E) Cell-cycle analysis by DNA content of LZ and DZ cells gated according to expression of CD86 and CXCR4. Left panel: DAPI profiles of cells in LZ and DZ gates. Middle panel: quantification of data from two independent experiments; each symbol represents one mouse. Right panel: cells in G2/M concentrate in the DZ gate. Percentages refer to G2/M cells.

p < 0.01; *p < 0.001; ****p < 0.0001, paired t test.

we depleted NP-specific cells from the GC by treatment with a large dose of NP-dextran, which strongly crosslinks the BCR of NP-specific cells. Consistent with the observation that BCR crosslinking results in GC B cell apoptosis (Han et al., 1995b; Pulendran et al., 1995; Shokat and Goodnow, 1995), injection of NP-dextran led to a marked loss of B1-8^{hi} cells from the GC (Figure S5C). This loss was followed by a shift in the DZ/LZ ratio of the endogenous cells to almost B1-8^{hi} levels (DZ/LZ ratio = 1.4:1), along with a robust increase in endogenous B cell proliferation and representation in the GC (Figure S5C). These data imply that migration of GC cells from LZ to DZ is controlled by competition for access to a limiting factor.

One such limiting factor could be the T helper cells present in the GC LZ, which might regulate selection on the basis of a B cell's ability to present cognate antigen (Allen et al., 2007a; Meyer-Hermann et al., 2006; Tarlinton, 2008). This model has been difficult to evaluate directly due to the lack of a suitable method for delivering T cell help without crosslinking the BCR. To isolate antigen capture from BCR crosslinking, we took advantage of the observation that GC B cells express high levels of DEC205, a cell-surface lectin that delivers antigen to MHC-containing processing compartments (Figure S5D) (Jiang et al., 1995; Kamphorst et al., 2010). Protein antigens such as OVA can be delivered to DEC205-expressing GC cells independently

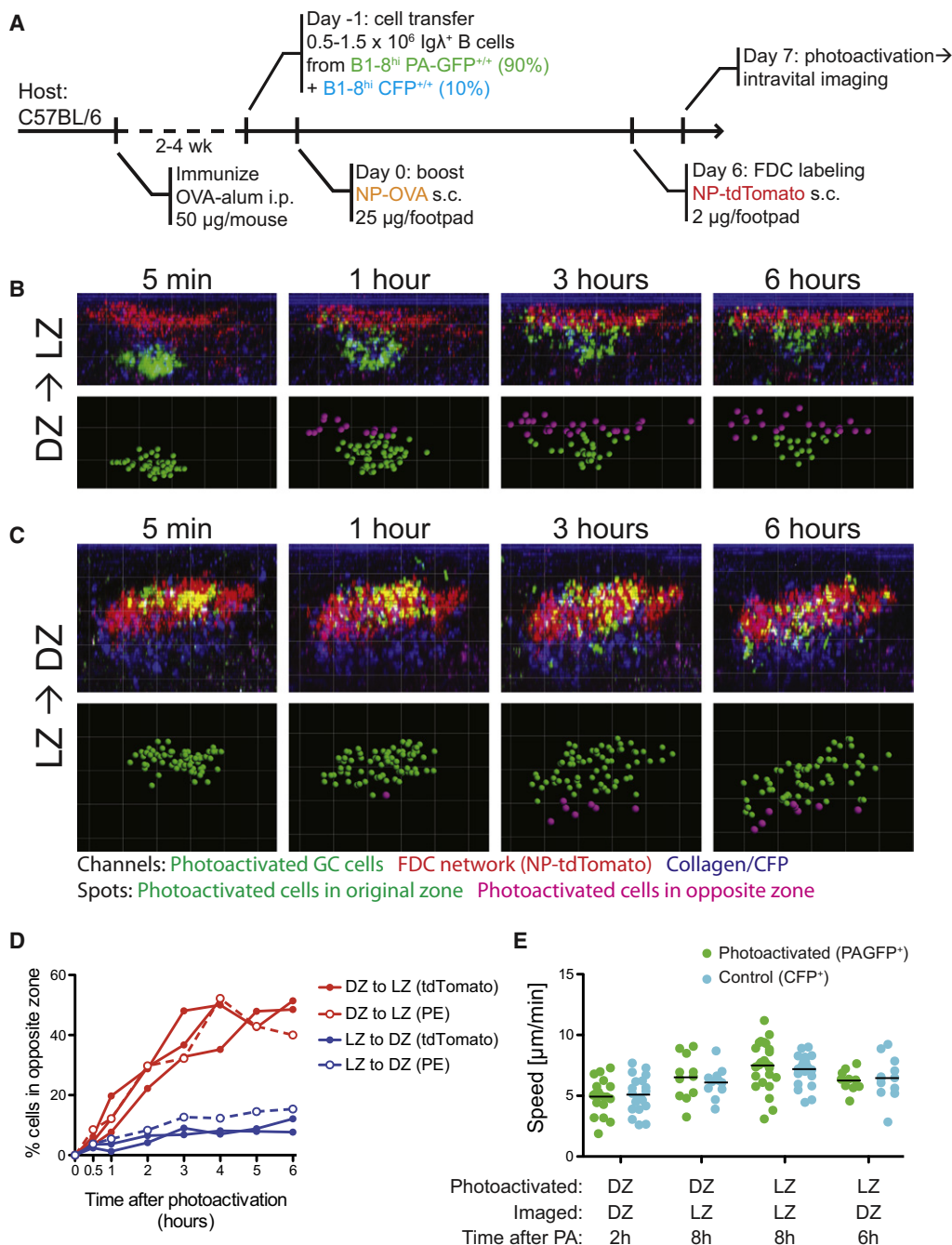


Figure 5. Interzonal Migration in GCs

(A) Diagrammatic representation of the experimental protocol.

(B) Time series showing position of individual cells photoactivated in the DZ, at different times after photoactivation. Top row: side (X-Z) view of whole GCs within popliteal lymph nodes of live mice. The LZ is labeled with NP-tdTomato (red); for better visualization, the PA-GFP signal shown (green) was isolated from autofluorescence and CFP bleed-through by colocalization filtering. Bottom row: spots were placed on photoactivated cells. Green spots indicate cells in the photoactivated zone, whereas magenta spots represent cells that have crossed over to the opposite (nonphotoactivated) zone.

(C) Time series for cells photoactivated in LZ. Details as in (B).

(D) Quantification of seven independent experiments (three LZ to DZ and four DZ to LZ, one of which was terminated at 4 hr). Closed symbols: LZ labeled by NP-tdTomato. Open symbols: LZ labeled by Phycoerythrin (PE) immune complexes.

(E) Speed of photoactivated GC cells and control CFP cells at different times after photoactivation. Each symbol represents one track; each column represents one cell type in one movie; bars represent the mean. Differences between PAGFP⁺ and CFP⁺ cells in the same movie were not significant ($p > 0.5$) by Mann-Whitney U test. Different movies were acquired using different settings and therefore should not be compared to each other.

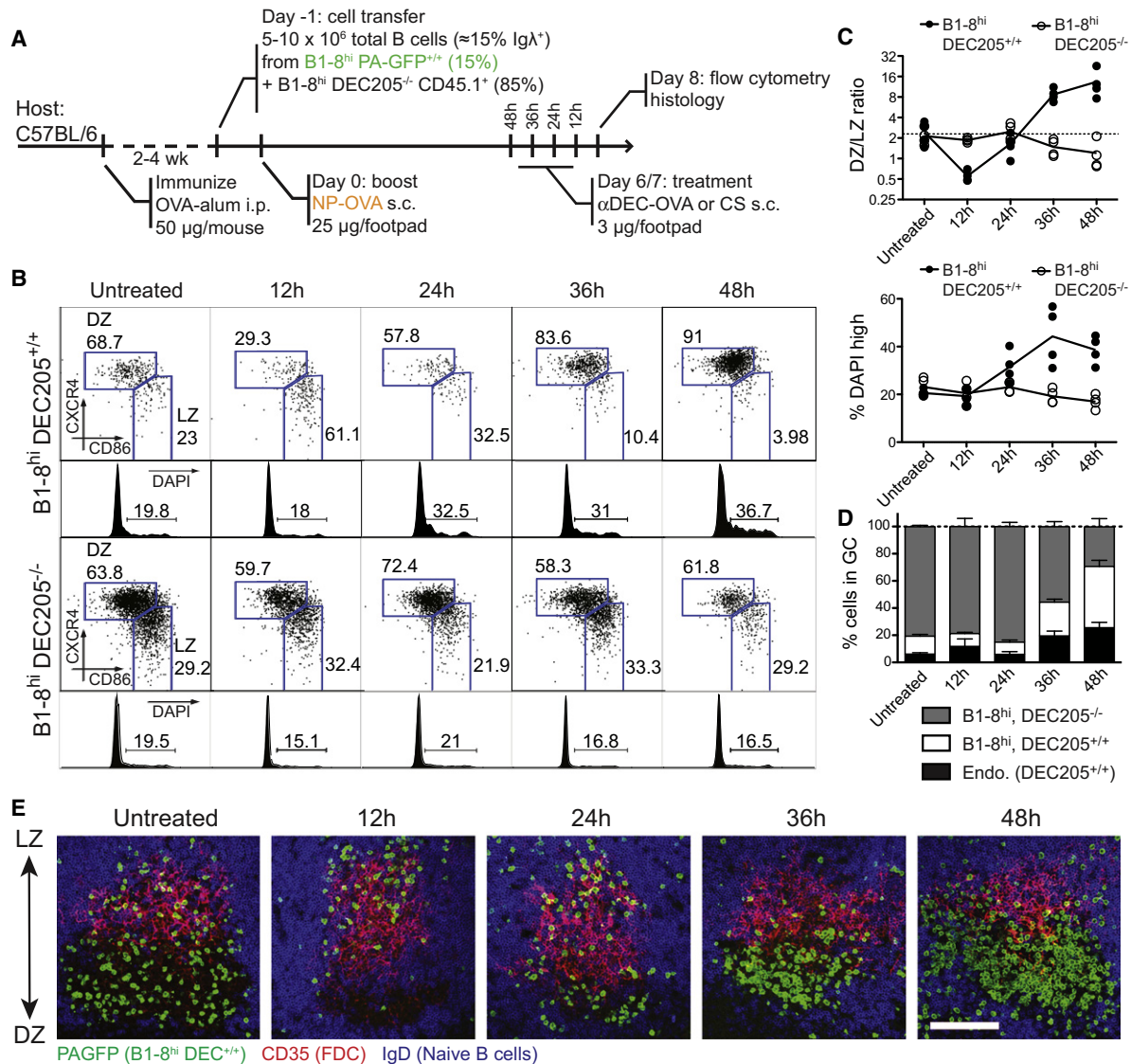


Figure 6. T Cell Help Availability Limits Interzonal Migration

(A) Diagrammatic representation of the experimental protocol.

(B) LZ/DZ distribution and DNA content (DAPI) analysis of DEC205^{+/+} (top) and DEC205^{-/-} (bottom) of popliteal lymph node GC B cells at different times after injection of αDEC-OVA.

(C) Quantification of (B). Each symbol represents one mouse.

(D) Fraction of GC cells derived from transferred DEC205^{+/+} and DEC205^{-/-} B1-8^{hi} and endogenous (DEC205^{+/+}) cells in popliteal lymph nodes of mice treated with αDEC-OVA for different lengths of time. Bars = SEM.

(E) Histology showing the distribution of DEC205^{+/+} cells in popliteal lymph node GCs at different times after injection of αDEC-OVA. DEC205^{+/+} cells, which express PA-GFP (used here as a genetic label), were stained with anti-GFP antibodies (green). LZ and DZ can be identified by presence of CD35^{hi} follicular dendritic cells (FDCs, in red) and absence of IgD⁺ B cells (in blue), respectively.

All data are representative of two experiments.

of BCR by using chimeric antibodies to DEC205 (αDEC-OVA). Targeting B cells in this manner results in efficient processing and presentation of OVA peptides (Figure S5E) (Kamphorst et al., 2010).

To test the role of antigen presentation in interzonal migration, we produced mixed GCs containing DEC205^{+/+} and DEC205^{-/-} B1-8^{hi} B cells by immunizing wild-type mice that had received

a mixture of the two cell types (Figure 6A). αDEC-OVA treatment initially resulted in a striking accumulation of DEC205^{+/+} cells in the LZ compartment (Figures 6B and 6C). However, by 24 hr after antigen delivery, DEC205^{+/+} cells began to reappear in the DZ, and by 48 hr, as much as 90% of this population was in the DZ (Figures 6B and 6C). Acquisition of DZ phenotype by DEC205^{+/+} cells was accompanied by a burst in proliferation

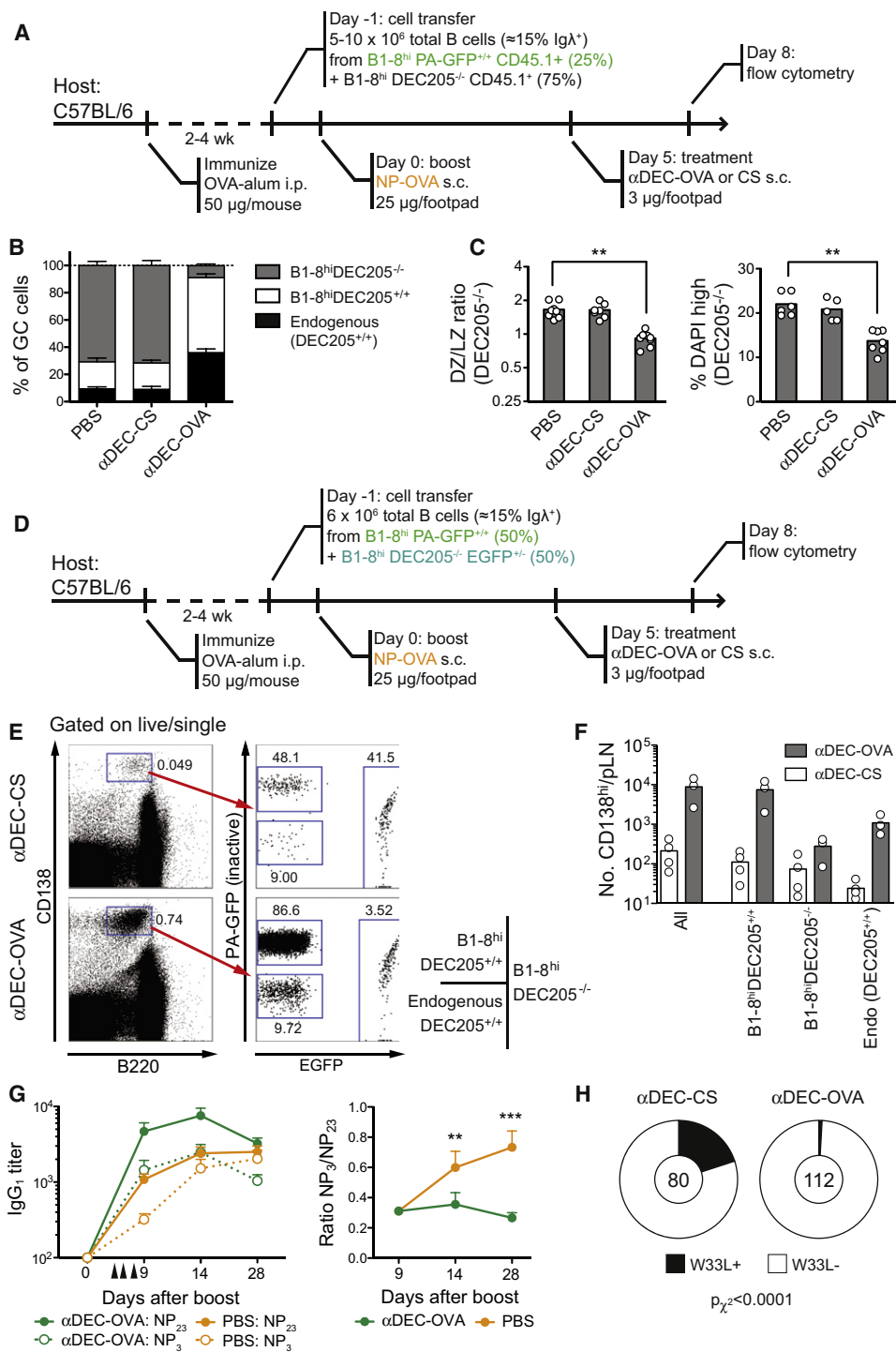


Figure 7. T Cell Help Limits Germinal Center Selection and Affinity Maturation

(A) Diagrammatic representation of the experimental protocol for (B)–(D). PA-GFP is used as a genetic label in its nonactivated form.

(B) Fraction of GC cells derived from transferred DEC205^{+/+} and DEC205^{-/-} B1-8^{hi} and endogenous (DEC205^{+/+}) cells in popliteal lymph nodes 72 hr after treatment with αDEC-OVA or αDEC-CS. Bars = SEM.

(C) DZ/LZ ratio and % cells in S/G2/M phase (DAPI high) among DEC205^{+/+} and DEC205^{-/-} popliteal lymph node GC B cells 72 hr after injection of αDEC-OVA or control αDEC-CS. **p < 0.01, Kruskal-Wallis test, Dunn's post-test.

(D) Diagrammatic representation of the experimental protocol for (E) and (F). PA-GFP is used as a genetic label in its nonactivated form.

(E) Plots show plasmablast/plasma cell gate (CD138^{hi}B220^{int}) and fraction of cells of each genotype, 72 hr after treatment with αDEC-OVA or αDEC-CS.

and a consequent increase in the relative number of these cells in the GC (Figures 6B–6D). Histological examination confirmed the relationship between expression of cell-surface markers and zonal localization (Figure 6E). In contrast, we found no alterations in GC B cell distribution or proliferation in DEC205^{−/−} cells in the same mice (Figures 6B–6D). Moreover, targeting with α DEC205 fused to an irrelevant antigen, *Plasmodium falciparum* circumsporozoite protein (α DEC-CS; Boscardin et al., 2006), had no effect on any GC B cell population (Figure S5F). Although there is no known cross-reactivity between the NP-specific B1-8^{hi} BCR and OVA, we examined the possibility of synergistic effects of DEC205 and BCR ligation by targeting GC B cells in NP-keyhole limpet hemocyanin (KLH) immunized mice with α DEC-OVA, as well as by co-crosslinking the two receptors with NP-conjugated α DEC205. Neither approach induced LZ to DZ migration or B cell expansion (Figure S5G), demonstrating that the effect of α DEC-OVA targeting is fully dependent on the specificity of GC T cells. We conclude that targeting available T cell help to a subpopulation of GC B cells is sufficient to induce en masse migration of this population from LZ to DZ, and that T cell help is therefore limiting for interzonal migration.

To determine whether T cell help is also limiting for clonal expansion, we examined mixed GCs containing both DEC205^{+/+} and DEC205^{−/−} B1-8^{hi} B cells 72 hr after treatment with α DEC-OVA (Figure 7A). Whereas control GCs targeted with an irrelevant antigen, α DEC-CS, maintained a ratio of DEC205^{+/+} to DEC205^{−/−} similar to the initial transfer (1:4), α DEC-OVA-targeted GCs were composed almost exclusively of DEC205^{+/+} cells (Figure 7B and Figure S6A). Importantly, the expansion of DEC205^{+/+} cells was accompanied by a decrease in the rate of cell division and skewing of the DZ/LZ ratio among the nontargeted (DEC205^{−/−}) population (from ~1.7:1 to 0.9:1; Figure 7C). The finding that the DEC205^{−/−} majority is counter-selected by focusing T cell help on a small subset of DEC205^{+/+} cells indicates that T cell help is indeed limiting for clonal expansion. Furthermore, α DEC-OVA treatment also led to a marked increase in the number of DEC205^{+/+} plasmablasts in targeted lymph nodes (Figures 7D–7F), showing that T cell help is also capable of regulating this other cell fate decision.

To determine whether restricting T cell help to high-affinity cells is a requirement for affinity maturation, we measured the effect of DEC205 targeting on the affinity of anti-NP serum antibodies in wild-type NP-OVA-immunized mice (Figure S6B). Because T cell help in targeted mice would be distributed among all GC cells irrespective of BCR affinity, we would expect such treatment to abrogate affinity maturation. As expected from the increase in plasmablasts in response to α DEC-OVA in the transfer experiments (Figures 7E and 7F), we found a significant increase in anti-NP titers in targeted mice (Figure 7G). However,

in contrast to control mice, the antibodies in α DEC-OVA-treated mice failed to undergo affinity maturation, as measured by the ratio of anti-NP₃ to anti-NP₂₃ titers (Figure 7G). Absence of affinity maturation in α DEC-OVA-targeted mice was also documented by their relative failure to accumulate the W33L mutation characteristic of high-affinity anti-NP antibodies (Allen et al., 1988), despite comparable overall mutation rates (Figure S6C, Figure 7H, and data not shown). Together, these experiments suggest that affinity maturation is abrogated when GC B cells receive T cell help independently of BCR affinity, and that, under conditions of equal T cell help, differential BCR crosslinking is not sufficient to mediate affinity-based selection. We conclude that T cell help is a limiting factor for B cell selection in GCs.

DISCUSSION

We used a combination of photoactivatable GFP, multiphoton microscopy, and flow cytometry to address long-standing questions regarding the mechanisms that govern selection in GCs. The results reveal distinct physiologic functions for the two anatomically defined GC compartments and suggest a cellular mechanism that limits affinity-based selection.

LZ and DZ Cells

The morphological and dynamic similarities between cells in the two GC zones revealed by intravital microscopy blurred the classical distinction between centrocytes and centroblasts (Allen et al., 2007b; Hauser et al., 2007a; Schwickert et al., 2007) to the extent that the existence of two distinct populations was called into question (Allen et al., 2007a). Two of the features traditionally thought to distinguish between these two populations were size—centroblasts being larger—and expression of surface immunoglobulin—thought to be higher in centrocytes (MacLennan, 1994). However, the size of B cells in the LZ and the level of surface immunoglobulin expression is virtually impossible to infer by histology (Allen et al., 2007a; Nieuwenhuis and Opstelten, 1984). Although highly desirable, clear distinction between the two cell types by flow cytometry was also not possible due to the lack of anatomically validated markers (Allen et al., 2007a; Högerkorp and Borrebaeck, 2006).

In situ photoactivation combined with flow cytometry resolves these problems by providing direct microanatomical information that allows for the unequivocal identification of LZ and DZ cells. In agreement with the initial multiphoton studies, we find that LZ and DZ cells are virtually equal in size (Allen et al., 2007b; Hauser et al., 2007a; Schwickert et al., 2007). However, despite their similarities, the two cell types differ in a number of important respects. Global gene expression analysis indicates that GCs are highly compartmentalized, with LZ cells showing evidence

(F) Absolute number of CD138^{hi}/B220^{int} of each genotype per popliteal lymph node.

(A–F) Each symbol represents one mouse; data pooled from 2–3 experiments.

(G) Titers of total (NP₂₃) and high-affinity (NP₃) NP-reactive IgG₁ (left) and total/high-affinity ratio (right) in NP-OVA-immunized wild-type mice treated or not with α DEC-OVA. Arrowheads indicate doses of treatment (details as in Figure S6B). Pooled data from two experiments, five mice per group per experiment. **p < 0.01; ***p < 0.001, Mann-Whitney U Test. Bars = SEM.

(H) Proportion of V186.2 clones bearing the high-affinity W33L mutation in day 14 wild-type GCs from mice treated with either α DEC-OVA or α DEC-CS (details as in Figure S6C). The number of clones analyzed is indicated in the center of each chart. Pooled data from two experiments, three mice per experiment.

of ongoing BCR and CD40 signaling activity consistent with antigen-dependent selection. In contrast, DZ cells specialize in cell division and express higher levels of Pol η , which mediates error-prone repair during somatic hypermutation (Delbos et al., 2005). Our data appear to refute recent claims that cell division occurs in both the LZ and the DZ (Allen et al., 2007b; Hauser et al., 2007a). However, the methods used in those studies labeled cells in the S phase of the cell cycle, and the results are therefore not incompatible with the idea that GC cells enter into S phase in the LZ, possibly in response to activation signals present in that zone, but leave this compartment before progressing to mitosis.

Interzonal Migration among GC B Cells

Multiphoton live-imaging studies provided the first concrete evidence for bidirectional GC B cell movement between zones (Allen et al., 2007b; Hauser et al., 2007a; Schwickert et al., 2007). However, due to limitations inherent to multiphoton microscopy (Allen et al., 2007a; Germain et al., 2006), the number of events was too small for robust statistical analysis (Figge et al., 2008). Moreover, true interzonal migrants could not always be distinguished from cells temporarily crossing the DZ/LZ border or exiting the GC from one zone while passing through the other. The combination of multiphoton imaging with PA-GFP allowed us to overcome these limitations by circumventing the need to track individual GC cells and enabled observation of the entire GC for periods of several hours. These longer-term experiments revealed polarized net movement of B cells from DZ to LZ, consistent with the idea that movement between zones supports iterative cycles of selection (Meyer-Hermann et al., 2001; Oprea and Perelson, 1997).

Mathematical modeling of our data using a simplified phenomenological model showed that roughly 30% of cells in the LZ are selected to return to the DZ. Though this model agrees with the experimental data to within one standard deviation, we expect that a more detailed model is needed to describe the complex process of interzonal migration. Particularly, it seems that saturation of the DZ to LZ migration frequency occurs faster in experiment than in theory. Although the reason for this is unclear, one possibility is that the DZ may comprise a heterogeneous population of cells with different migration properties (e.g., pre- and postmitotic cells), or that a proportion of DZ cells may be arrested in this zone due to genetic errors generated by somatic hypermutation.

Interaction with T Cells Is Limiting for Affinity-Based Selection

Selection of high-affinity B cells in GCs is believed to be mediated by limiting amounts of antigen on the surface of FDCs (Eisen and Siskind, 1964; Goidl et al., 1968; Nussenzweig and Benacerraf, 1967). It has been proposed that this effect is governed by direct signals derived from BCR crosslinking (Brink et al., 2008; Liu et al., 1989; Manser, 2004; Paus et al., 2006; Phan et al., 2006; Tarlinton and Smith, 2000). However, in addition to signaling, the BCR is also an endocytic receptor that captures antigen for processing and presentation to T cells (Lanzavecchia, 1985; Rock et al., 1984). Thus, an alternative, though nonexclusive, model for selection in the GC is that it is mediated

by helper T cells, which recognize peptide-MHC presented on the surface of high-affinity B cells (Allen et al., 2007a; Meyer-Hermann et al., 2006; Tarlinton, 2008; Vinuesa et al., 2005b). The role of the BCR in this model would be to retrieve antigen from the FDC surface, with affinity dictating the relative amount of antigen captured (Batista et al., 2001; Suzuki et al., 2009).

Although it is clear that both BCR and T cell-delivered signals are necessary for GC maintenance (Han et al., 1995a; Huntington et al., 2006), and that each of these signals is capable of extending B cell survival in vitro (Liu et al., 1989), evidence for one or the other as agents of selection in the GC is circumstantial (Allen et al., 2007a; Tarlinton, 2008). Importantly, no experimental models to our knowledge have been able to acutely induce the selection of a specific GC population in vivo. Our experiments show that helper T cells can control interzonal migration by directing B cells that capture antigen to return from the LZ to the DZ, where they undergo clonal expansion. Similarly, helper T cells also appear to regulate a B cell's decision to develop into an antibody-secreting plasmablast (Figures 7F and 7G). The finding that targeting T cell help to GC B cells leads to their selective expansion and differentiation at the expense of nontargeted cells implies that T cell help is limiting for GC selection. This notion is supported by previous work showing that dysregulated T cell help can lead to faulty GC selection and to the emergence of autoantibody-producing B cells (Vinuesa et al., 2005a; Kim et al., 2010). Although our experiments do not rule out a role for BCR crosslinking in selection, they suggest that antigen receptor crosslinking is not sufficient for selection when B cells with receptors of different affinities display equivalent levels of peptide-MHC (Figures 7G and 7H). Under physiological conditions, it is likely that signals delivered by BCR crosslinking and T cell help interact with additional input from FDCs, Fc receptors, and toll-like receptors to facilitate affinity maturation.

A Dynamic Model of GC Selection

The exquisite dependence of T cell-induced selection on migration between zones shown in our experiments suggests a highly polarized GC structure, with different compartments specialized in selection and proliferation. We propose a model wherein signals from antigen in the LZ initially entrap specific B cells in this zone. Subsequent presentation of peptide-MHC to follicular helper T cells produces signals that eventually lead to B cell migration to the DZ. Although both high- and low-affinity B cells can capture antigen, the amount of antigen captured will be directly proportional to the affinity of their BCRs, leading to an affinity-dependent difference in the amount of processing and presentation. A limiting number of T cells then selects those B cells with the highest peptide-MHC density (Depoil et al., 2005) and directs their return to the DZ, where they undergo rapid division. Conversely, LZ cells that fail to be selected are relatively enriched in the LZ, where they eventually undergo apoptosis—as suggested by the enhanced expression of proapoptotic genes in this compartment. Although affinity maturation in this model does not require the active selection of high-affinity clones into the plasma cell fate, it seems that this decision too can be regulated by the same mechanism. We conclude that T helper cells limit the selection of GC B cells by triggering their return from the LZ to the DZ in a functionally compartmentalized GC.

Although transgenic mice expressing photoconvertible fluorescent proteins have been reported previously (Nowotschin and Hadjantonakis, 2009; Tomura et al., 2008), this is the first time, to our knowledge, that transgenic PA-GFP has been combined with multiphoton microscopy and flow cytometry to allow precise labeling, tracking, and molecular analysis of cells deep within living tissues. This powerful combination of tools is likely to be applicable to problems in other areas of biology.

EXPERIMENTAL PROCEDURES

Mice

PA-GFP-transgenic mice were generated by cloning the PA-GFP ORF (Patterson and Lippincott-Schwartz, 2002) downstream of the human Ubiquitin C promoter (Schaefer et al., 2001). A linearized 2.7 Kb fragment containing the transgene was injected into C57BL/6 (B6) fertilized female pronuclei. Wild-type B6, B6.SJL, and B10.BR mice, as well as B6 mice ubiquitously expressing ECFP, dsRed, or EGFP, were purchased from Jackson Laboratories. B1-8^{hi} knockin (Shih et al., 2002) and DEC205^{-/-} mice (Guo et al., 2000) were generated and maintained in our laboratory.

Photoactivation and In Vivo Imaging

Cells were photoactivated by scanning with a femtosecond-pulsed multiphoton laser tuned to 830 nm wavelength and then imaged at 940 nm wavelength. Conditions for photoactivation and intravital imaging are detailed in the Extended Experimental Procedures.

Image Analysis

Cell position in interzonal migration time courses was scored manually in 3D using Imaris software v. 6.4.0, as detailed in the Extended Experimental Procedures. Cells were only considered as being in the zone opposite to that of photoactivation after having crossed a transitional area of weak FDC staining.

In Situ Labeling for Flow Cytometry and Cell Sorting

Carefully dissected lymph nodes were placed between two microscope coverslips, held apart using vacuum grease. For each individual lymph node, either DZ or LZ was photoactivated. When possible, LZ and DZ were photoactivated in contralateral nodes from the same mouse. Individual nodes were processed immediately after photoactivation, and pools of 3–5 nodes per condition were stained for flow cytometry with the reagents listed in Table S4. For microarray experiments, samples were kept on ice or at 4°C during all stages from dissection to sorting.

Microarray

Photoactivated LZ or DZ GC cells were sorted directly into Trizol reagent (Invitrogen). RNA from pooled samples from 15–20 lymph nodes containing at least 1.2×10^4 cells was subjected to two-round amplification using the Affymetrix 3' IVT Express Kit. cDNA was hybridized to GeneChip MOE430 2.0 arrays (Affymetrix) according to the manufacturer's instructions. RNA extraction, amplification, and hybridization were carried out at the Memorial Sloan Kettering Cancer Center (MSKCC) Genomics Facility.

qPCR

LZ and DZ cells were sorted directly into Trizol LS reagent (Invitrogen) and RNA extraction was carried out according to the instructions provided by the manufacturer. RNA was reverse-transcribed using random hexamers and SuperscriptII reverse transcriptase (Invitrogen). Reverse-transcribed RNA was amplified using the primers presented in Table S5 and SYBR Green PCR mix (Applied Biosciences). All primers were designed to span at least one intron. Expression levels were normalized to *Gapdh* and fold changes were calculated by dividing normalized expression in LZ by DZ.

Anti-DEC205 Treatment

α DEC-OVA, α DEC-CS, α DEC-HEL_{48–62}, and ISO-HEL_{48–62} were produced as chimeric antibodies in 293T cells as described (Boscardin et al., 2006) and injected into mice with ongoing GCs as detailed for each experiment.

Statistical Analysis

Comparisons between different populations within the same sample or between pairs of samples from the same experiment were done by paired t test. Other pair comparisons were done by Mann-Whitney U test. Comparisons between three groups were done by Kruskal-Wallis test with Dunn's multiple comparison post-test. The proportion of high-affinity sequences in α DEC-OVA- and α DEC-CS-treated mice was compared by χ^2 test with Yates correction. All analyses were carried out using Prism software v. 5.0 (Graphpad).

ACCESSION NUMBERS

The GEO accession number for the microarray data reported in this paper is GSM589872.

SUPPLEMENTAL INFORMATION

Supplemental Information includes Extended Experimental Procedures, six figures, five tables, and five movies and can be found with this article online at doi:10.1016/j.cell.2010.10.032.

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